

LFER Part 4: Steric Parameters

Barry Linkletter

We have seen a two-parameter plot in the Yukawa-Tsuno equation that separated inductive and resonance electronic effects of substituents. We will now explore a two-parameter system that separates electronic and steric effects as we consider the Taft equation. We will further define other common LFER schemes used for interpreting reaction mechanisms.¹

The Taft Plot and More

The Hammett plot is a cheat.² The steric demands are held constant by the same phenyl ring being near the reaction centre while the changing substituents are held on the far side. Robert Taft introduced a two-parameter system³ for more "real-world" systems where the substituent is not isolated from the reaction centre.^{4,5}

1. Plot the data for acid hydrolysis of substituted ethyl benzoates⁶ presented in table 1 against the Hammett σ values.
 - (a) What is the reaction constant, ρ , for acid-catalyzed ester hydrolysis? Is there an electronic effect in this process?
 - (b) Write out the mechanism for acid and base-catalyzed hydrolysis of an ester. Use these mechanism to explain the observed reaction constants, ρ , in each case.
 - (c) Use the values for the acid hydrolysis of substituted ethyl acetates in table 3 to determine your own Taft E_s steric parameters for CH_2Cl , CHCl_2 and CCl_3 (set $E_s = 0$ for the CH_3 group, just like Taft did.) Compare them to the literature values presented in table 5)
 - (d) Taft used the reaction constant for base-catalyzed hydrolysis of ethyl benzoates as the "standard ρ value". Plot data for such a reaction and determine the reaction constant for this reaction. Table 2 collects some rate data from Ingold⁷ and Evans.⁸
 - (e) Using the values above, Taft and other established a list of substituent parameters, σ^* for groups attached to the carbonyl of an ester. Are these generally applicable in aliphatic systems? Plot the pK_a values for various carboxylic acids and alcohols vs. the Taft σ^* values. See table 4 and table 5 for data.
2. In your reading of the textbook you will have encountered the following LFER concepts. Answer the following questions to help yourself pay closer attention to them.

This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were created in ChemDoodle Plots were typeset using Matplotlib and Python tools. Diagrams were created and edited using Affinity Designer.

¹ Read chapter 8.3 and 8.4.

Figure 1: Substituted ethyl benzoate and ethyl acetate (when $\text{X} = \text{CH}_3$)

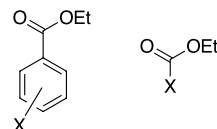


Figure 2: The Taft equation

$$\log \frac{k_X}{k_{\text{CH}_3}} = \rho^* \sigma^* + \delta E_s$$

² Sorry to burst your bubble. Also, "the cake is a lie!"

³ See ch. 8.4.1

⁴ "Polar and Steric Substituent Constants for Aliphatic and o-Benzoate Groups from Rates of Esterification and Hydrolysis of Esters." R.W. Taft, *J. Am. Chem. Soc.*, 1952, 74, 3120-3128. <https://doi.org/10.1021/ja01132a049>

⁵ "Linear Free Energy Relationships from Rates of Esterification and Hydrolysis of Aliphatic and Ortho-substituted Benzoate Esters." R.W. Taft, *J. Am. Chem. Soc.*, 1952, 74, 2729-2732. <http://doi.org/10.1021/ja01131a010>

⁶ "The activation energy of organic reactions. Part III. The kinetics of acid hydrolysis of esters." E.W. Timm, C.N. Hinshelwood, *J. Chem. Soc.*, 1938, 168, 862-869. <http://doi.org/10.1039/jr9380000862>

⁷ "Mechanism of, and constitutional factors controlling, the hydrolysis of carboxylic esters. Part VIII. Energies associated with induced polar effects in the hydrolysis of substituted benzoic esters." K.C. Ingold, W.S. Nathan, *J. Chem. Soc.*, 1936, 222-225. <https://doi.org/10.1039/jr9360000222>

⁸ "Studies of the ortho-effect. Part III. Alkaline hydrolysis of benzoic esters." D.P. Evans, J.J. Gordon, H.B. Watson, *J. Chem. Soc.*, 1937, 1430-1432. <https://doi.org/10.1039/jr9370001430>

- write out the Grunwald-Winstein equation. Define the terms and be able to describe the reference reaction. How are Grunwald-Winstein solvent parameters, ρ , useful?
- Write out the Swain-Scott equation. Define the terms and be able to describe the reference reaction. How are Swain-Scott nucleophilicity parameters, n_x , useful?
- Explain how the Brønsted plot relates nucleophilicity and leaving group ability to the pK_a values of the conjugate acid of the nucleophile or leaving group. How are Brønsted plots useful? What are the limitations of this approach?

Data Tables

Substituent ⁸	$k / 10^{-5} \frac{\text{mol}}{\text{Ls}}$	Substituent ⁷	$k / 10^{-5} \frac{\text{mol}}{\text{Ls}}$
p-NH ₂	1.27	H	62.1
p-OCH ₃	11.5	m-CH ₃	43.3
p-CH ₃	25.1	m-Cl	477
H	55.0	m-NO ₂	4290
p-Cl	237	p-F	126
p-I	278	p-NO ₂	7200
p-Br	289		
p-NO ₂	5670		

Substituent (X)	pK_a (XCOOH)	pK_a (XCH ₂ OH)
CH ₃	4.76	16
CH ₂ Cl	2.86	14.31
CH ₂ Br	2.86	
CHCl ₂	1.29	12.89
CCl ₃	0.65	12.24
CBr ₃	0.66	
CF ₃	0	12.37
CH ₂ F	2.66	
C ₆ H ₅	4.31	15.4
CH ₂ CN	2.43	
CH ₂ OCH ₃	3.53	
C ₂ F ₅		11.35

Table 1: Rate of acid-catalyzed solvolysis for ethyl benzoates in 50% ethanol, 100 °C.⁶

Substituent	$k / 10^{-5} \text{ s}^{-1}$
p-OCH ₃	6.17
p-OH	4.47
p-CH ₃	7.76
H	9.12
p-Cl	7.94
p-Br	8.31
p-NO ₂	10.7
m-NO ₂	8.91
o-NO ₂	0.513

Table 2: Rate of base-catalyzed solvolysis for ethyl benzoates in 85% ethanol, 25 °C.^{7,8}

Table 3: Rate of acid-catalyzed solvolysis for ethyl acetates in 50% ethanol, 100 °C.⁶

Substituent	$k / 10^{-5} \text{ s}^{-1}$
CH ₃	1072
CH ₂ Cl	724
CHCl ₂	275
CCl ₃	550

Table 4: pK_a values for selected carboxylic acids and alcohols, pK_a values obtained from the Williams pK_a compilation [PDF]

Table 5: Taft E_s steric and σ^* electronic parameters

Substituent (X)	E_s	σ^*
Me	0	0
Et	-0.07	-0.1
Pr	-0.36	-0.115
i-Pr	-0.47	-0.19
n-Bu	-0.39	-0.13
i-Bu	-0.93	-0.21
t-Bu	-1.54	-0.30
Et ₃ C	-3.8	
ClCH ₂	-0.24	1.05
Cl ₂ CH	-1.54	1.94
CCl ₃	-2.06	2.65
BrCH ₂	-0.27	1.00
Br ₂ CH	-1.86	1.96
CBr ₃	-2.43	2.43
CH ₂ F	-0.24	1.10
CF ₃	-1.16	2.61
CH ₂ C ₆ H ₅	-0.38	0.215
CH ₂ CN	-0.94	1.30
CH ₂ OCH ₃		0.52
C ₂ F ₅		2.56
C ₆ H ₅	-2.48	0.60

from "Free Energy Relationships in Organic and Bio-organic Chemistry" by A. Williams, 2003, pg 272. <https://doi.org/10.1039/9781847550927>.

Using Steric Parameters to Improve a Reaction

An example of an organometallic reaction that makes a carbon-carbon bond with a high degree of stereoselectivity.¹³ We will see how a group of researchers used Taft steric parameters to investigate and improve the catalyst.

Grignard and Grignard-like reactions

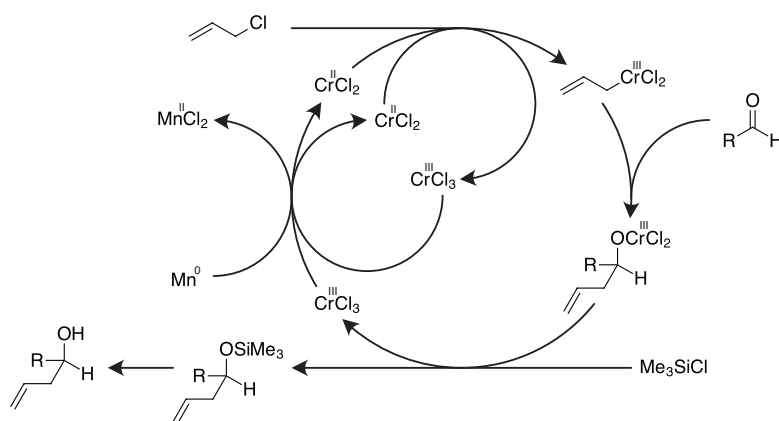
You are familiar with the classic Grignard reaction. We make a metalated carbon nucleophile via oxidation of a metal like zinc. We then add the nucleophilic carbon group to a solution containing an electrophile such as a ketone or aldehyde. The result is a new carbon-carbon bond.

The Grignard reaction requires dry conditions and aprotic solvents. The Barbier reaction is similar but the zinc oxidation takes place *in situ* with the two carbon reactants. This reaction is insensitive to water and have been carried out in water solvent, making it a candidate for "green chemistry."⁹

The Nozaki-Hiyama-Kishi (NHK) reaction

The NHK reaction¹⁰ uses chromium to enable the addition of an allyl bromide to a carbonyl group. The dirty little secret of all Grignard-like chemistry is that they are really organometallic reactions where the metal hides a radical mechanism.

A recent¹¹ example of a modified NHK reaction uses chiral ligands to provide stereoselectivity¹² for the addition reaction and features the use of steric LFER parameters to investigate the reaction.¹³ Read this contribution and consider the following questions and discussion. This is an impressive result. By including a chiral ligand with the chromium salt the author achieved a high degree of stereoselectivity.



The catalytic cycle in figure 5 outlines a hypothesis for the mechanism of this version of the NHK reaction. The allyl bromide¹⁴ is metalated via

Figure 3: A Grignard reaction.

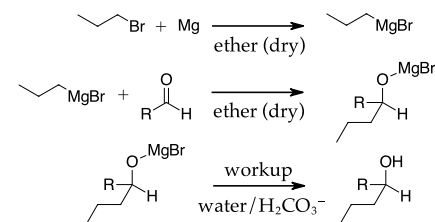
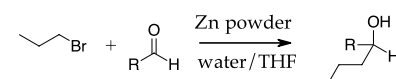


Figure 4: A Barbier reaction.



⁹ Here is an example of a Barbier reaction performed using no solvent – very "green" indeed: "A Mechanochemical Zinc-Mediated Barbier-Type Alkylation Reaction under Ball-Milling Conditions." J. Yin et al., *J. Org. Chem.*, 2020, 85, 2347–2354. <https://doi.org/10.1021/acs.joc.9b02876>

¹⁰ For a useful dictionary of famous named reactions in chemistry see "Name reactions and reagents in organic synthesis" by B.P. Mundy, M.G. Ellerd and F.G. Favaloro, *J. Wiley & Sons, Hoboken, N.J.*, 2005, ISBN 0-471-22854-0. It is available in the UPEI library.

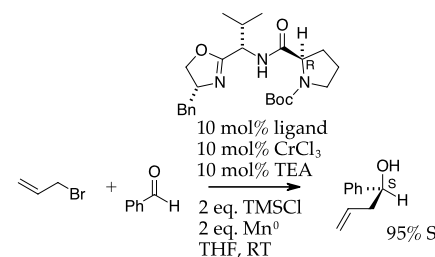
¹¹ The word "recent" in physical organic chemistry means "this century."

¹² Chapter 6 of the textbook discusses stereochemistry in detail.

Figure 5: A mechanism for the Nozaki-Hiyama-Kishi reaction

¹³ "Examination of the Role of Taft-Type Steric Parameters in Asymmetric Catalysis." M.S. Sigman, J.J. Miller, *J. Org. Chem.*, 2009, 74, 7633–7643. <https://doi.org/10.1021/jo901698t>

Figure 6: A stereoselective Nozaki-Hiyama-Kishi reaction.¹³



¹⁴ I presented allyl chloride in figure 5 for simplicity. The allyl bromide mechanism is the same but you will have a mixture of bromide and chloride counterions.

the oxidation of 2 eq. of Cr^{II} to Cr^{III} . The Cr^{III} is reduced by manganese metal to give the Mn^{II} salt byproduct and regenerate Cr^{II} so the cycle can continue. The trimethylsilyl chloride (TMSCl) is used to separate the alkoxide product from the Cr^{III} salt so that the salt is liberated to participate in the catalytic cycle. The initial silyl ether product is cleaved to give the alcohol during the workup procedure. TMS is the most sensitive of the silyl protecting groups.¹⁵ Although the deprotection reaction is not described in the procedure,¹³ the conditions of the workup are what cleaves the silyl ether. It would have been nice if the authors had discussed this in the paper but perhaps those familiar with the craft deem it obvious.

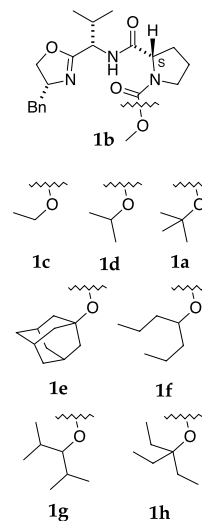
The Ligand Series

The ligand is chelated to the chromium centre and favours one enantiomeric product over the other during the addition reaction. The planar aldehyde group can approach the ligand/metalated carbon centre with either the *re* or *si* face. If we were to use the *R* ligand as shown in figure 7 then we have two diastereomeric transition states possible: the *R:re* and *R:si* options. If the energy difference between the diastereomers is about $2.5 \frac{\text{J}}{\text{mol}}$ then that would give you 95 % or one enantiomeric product over the other.

The authors chose a series of ligands with increasing bulk on the proline group and plotted the log of the enantiomeric ratio (*R*/*s*) vs. an LFER parameter for steric bulk.¹³ They chose to use Charton parameters, v ,¹⁶ which are a calculated series based on the Van der Waals radius of the attached atoms. I plan to plot the the results against Taft parameters, E_s , for the substituents in my own analysis of the results. Taft parameters are based on experimental observations, unlike Charton parameters that are based on mathematical calculations of atomic volumes. The authors of the paper make an argument that the Charton approach is superior, but I wonder if that is just because those parameters gave them better correlations.

¹⁵ "Selective Deprotection of Silyl Ethers." T.D. Nelson, R.D. Crouch, *Synthesis*, 1996, 1031-1069. <https://doi.org/10.1055/s-1996-4350> or "Greene's Protective Groups in Organic Synthesis" by P.G.M. Wuts, T.W. Greene, 2014, John Wiley & Sons, Inc., <https://doi.org/10.1002/9781118905074>. Available via the UPEI library.

Figure 7: The ligand series explored in the paper.¹³



¹⁶ for an explanation and tables of v parameters see "The upsilon steric parameter – definition and determination." M. Charton, *Topics Curr. Chem.*, 1983, 114, 57-91. <https://doi.org/10.1007/BFb0111213>

Table 6: Experimental data reported in the paper.¹³

Ligand	Group	Substituent	E_s	E'_s ¹⁷	v_x ¹⁶	%R using this Electrophile ¹³		
						Ph(CO)H	Ph(CH ₂) ₂ (CO)H	Ph(CO)CH ₃
1a	t-Bu	C(CH ₃) ₃	−1.54	−1.43	1.24	96	74	69
1b	Me	CH ₃	0	0	0.52	60	51	25
1c	Et	C ₂ H ₅	−0.07	−0.06	0.56	65.5	52.5	24
1d	i-Pr	CH(CH ₃) ₂	−0.47	−0.48	0.76	82.5	56.5	37.5
1e	1-adamantyl				1.33	96	73.5	72
1f	n-Pr ₂ CH	CH(C ₃ H ₇) ₂	−2.11	−2.03	1.54	73		37.5
1g	i-Pr ₂ CH	CH(CH(CH ₃) ₂) ₂		−5.01	1.70	77.5		55
1h	Et ₃ C	C(C ₂ H ₅) ₃	−3.8	−5.29	2.38	91.5		61.5

Let us do our own analysis of the data. We can now plot the $\log R/s$ vs. E_s ¹⁷ and see if we get a similar correlation between enantiomeric ratio and size parameter as the authors.¹⁸

Data Analysis – Charton Plot

Before we do that, let us plot the data using the Charton steric parameter, v , as the authors did in figure 5 of the paper¹³ (reproduced in figure 8 of this document.) Take note of this argument that there are two regimes of steric sensitivity. The Taft plot has a slope of 1.5 for the substituents with less bulk and then the stereoselectivity plummets and a new regime begins with a slope of 0.7.

I find this argument unsatisfying. The authors argue that steric strain favours one enantiomer over the other until a limit is reached where a new catalyst structure occurs and then increased bulk again begins improving the stereoselectivity. There is clearly a trend of increasing stereoselectivity until a threshold is reached where the structures of the catalysts change in some manner. However, I do not find three-point plots very compelling. I favour the simpler hypothesis that increased steric bulk provides improves stereoselectivity until we go too far and it all goes to pieces. I do observe that, in the case of acetophenone, the authors did not propose a line fit for those three points.

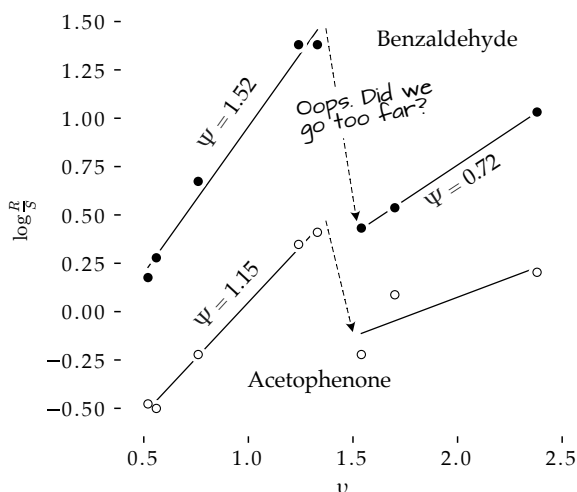


Figure 8: Plot of the log of the enantiomeric ratio vs. Charton steric parameters for the allylation of benzaldehyde and acetophenone as presented in figure 5 of the original paper.¹³ Experimental data is shown in table 6.

Data Analysis – Taft Plot

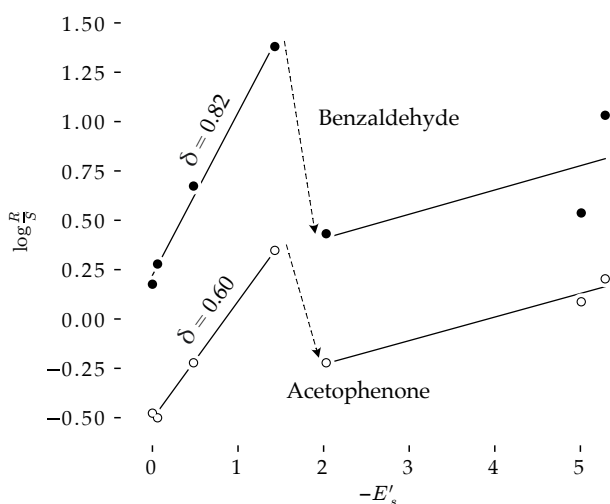
Charton steric parameters are designed to express the steric volume of the atoms closest to where the group is connected.¹⁶ Taft steric parameters are based on experimental observations and provide a "real world" measure-

¹⁷ I will be using the improved Taft values, E_s' , developed by Dubois et al. See "Steric effects—I: A critical examination of the taft steric parameter— E_s . Definition of a revised, broader and homogeneous scale. Extension to highly congested alkyl groups." J.A. MacPhee, A. Panaye, J.-E. Dubois, *Tetrahedron*, 1978, 34, 3553–3562. [https://doi.org/10.1016/0040-4020\(78\)88431-2](https://doi.org/10.1016/0040-4020(78)88431-2)

¹⁸ See the interactive Python notebook that accompanies this document for documentation of the calculations and plotting.

ment of the steric effect of a group in acid-catalyzed ester hydrolysis.¹⁷ The two parameter sets track well together but there are exceptions.

In figure 9 I have plotted the same enantiomeric ratio data against Taft E'_s . The results are similar.¹⁹ The Taft E'_s list of parameters does not have a value for the adamantyl group and the values are more extreme for the bulkiest groups. Observe that the correlation in the second set of three points with the highest steric bulk is no longer telegraphing a straight line. The general conclusion remains the same – there is a direct correlation with steric parameters until a threshold is reached.



¹⁹ See the interactive Python notebook that accompanies this document for documentation of the calculations and plotting.

Figure 9: Plot of the log of the enantiomeric ratio vs. Taft steric parameters for the allylation of benzaldehyde and acetophenone. Experimental data is shown in table 6.

Data Analysis – Comparing E'_s and v

The authors of the paper were able to get a straighter line for the three-point plots after the break in the trend using Charton v parameters compared with Taft E'_s parameters. Perhaps this is why the author preferred Charton to Taft?

We can plot the data set of E'_s and v against each other as shown in figure 10.²⁰ There are many substituents that have values for both parameters. The result is encouraging and disappointing. We see a clear and almost unshakeable correlation between the Taft E'_s and Charton v parameters with a handful of very obvious exceptions. The calculated Charton parameters and the experimental Taft parameters match up almost perfectly. However for very bulky substituents we see some deviations. These deviations however are inconsistent. Some bulky groups match the correlation while several not-too-dissimilar groups do not. Unfortunately for this study, two of the bulkiest groups used in the data are ones that fall far off the line.

²⁰ See the interactive Python notebook that accompanies this document for documentation of the calculations and plotting.

The most extreme Taft steric parameters are part of the data set produced by Dublois et al.¹⁷ and those authors did express their surprise at the high magnitude of the values. They did, however, assert their accuracy in their discussion. These values were obtained by competition experiments against other substituents so there is the possibility of increased error. Perhaps they are just wrong?²¹

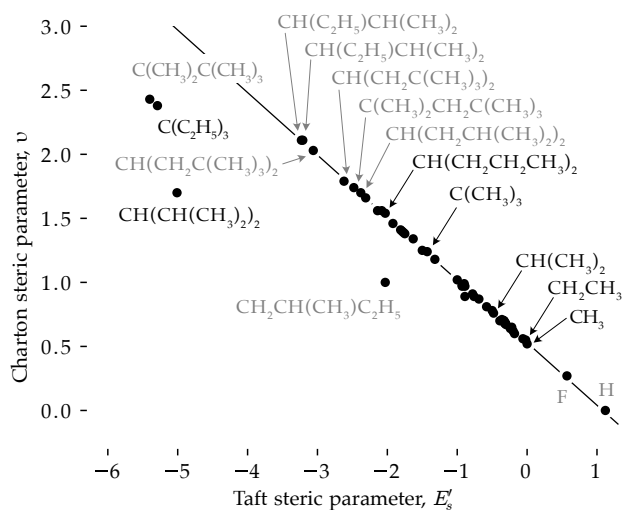


Figure 10: A plot of the correlation between the Taft E'_s and the Charton v steric parameters. The black substituents are used in the plots here, the grey substituents are shown for comparison.

²¹ For more on the efforts of Dubois et al. to confirm and justify their Taft steric parameters for bulky systems see: "Steric effects—II: Relationship between topology and the steric parameter. E'_s – topology as a tool for the correlation and prediction of steric effects." A. Panaye, J.A. MacPhee, J.-E. Dubois, *Tetrahedron*, 1980, 36, 759–768, [https://doi.org/10.1016/S0040-4020\(01\)93691-9](https://doi.org/10.1016/S0040-4020(01)93691-9);

"Steric effects—III: Composition of the E'_s parameter. Variation of alkyl steric effects with substitution. Role of conformation in determining sterically active and inactive sites." J.-E. Dubois, J.A. MacPhee, A. Panaye, *Tetrahedron*, 1980, 36, 919–928, [https://doi.org/10.1016/0040-4020\(80\)80043-3](https://doi.org/10.1016/0040-4020(80)80043-3);

"Steric effects. 4. Multiparameter correlation models. Geometrical and proximity site effects for carboxylic acid esterification and related reactions." J.A. MacPhee, A. Panaye, J.-E. Dubois, *J. Org. Chem.*, 1980, 45, 1164–1166. <https://doi.org/10.1021/jo01294a052>

One benefit of the Charton system is that one does not need to scour the literature looking for kinetic data of sterically-strained acid-catalyzed ester hydrolysis, adjust the values for temperature and reaction conditions and then use that experimental result to calculate a Taft E'_s value for that given group. One just uses a calculator to produce the parameter based on Van der Waals radii and Charton's recipe.

Conclusion

I began this discussion with a plan to show that the Taft E'_s were just as good for analyzing the organometallic stereoselective reaction investigated by the authors as the Charton v parameters. In the end I have come full-circle and now accept that the Charton values were chosen for a good reason. The authors did not trust the E'_s for bulky substituents. Also, the Charton method can be used to produce steric parameters for just about any group.

One soothing fact is that groups for which the Taft and Charton values correlate include the first point after the break in the trend. So the enantioselectivity does increase with steric bulk and then the catalyst "breaks" when the substituents are too bulky.

I still do not agree with declaring a correlation after the break without better information about the structural change. Just because the shattered

remnants fell in a straight line does not guarantee that it is a real trend.

Future Work

I see a gap between substituents with $E'_s = -1.43$ (t-Bu) and $E'_s = -2.0$ (CH(Pr)₂) that should be explored. somewhere in there may be even better enantioselectivity before it goes to ruin. What substituents have an E'_s value around 1.6 and 1.8?

Textbook Questions

As you read about the Taft equation and other linear free-energy relationships, try these textbook problems: Ch. 8: 9, 10, 12.

Other Questions

1. Show that the enantiomeric ratio, R/S , is a measure of free energy difference and is suitable for using in an LFER plot.
2. Why did the authors not include any electronic effects in their consideration of substituent effects?
3. Propose a ligand that would potentially have a better R/S ratio for product than ligand **1a**. Justify your choice of substituent. Now propose a synthesis of this ligand. Include all reaction conditions in your plan.²²

²² There might be a detailed description of the synthesis of **1a** out there somewhere that you could use as a guide. I wonder where one might find it?

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have learned about steric parameters and how they are defined in the Taft equation.
- be able to define the Taft σ^* parameters for electronic effects and describe how they compare to the corresponding Hammett σ values.
- have used experimental data to create your own Taft E_s parameters.
- be able to define and describe other LFER systems such as the Grunwald-Winstein equation, the Swain-Scott equation and the Brønsted plot.
- have experienced synthetic planning for a target molecule.
- **Computer Skill:** be able to use an interactive *Python* notebook to interpret and analyze experimental data from the literature.
- **Computer Skill:** be able to use a text editor to create data sets using experimental data from the literature and make these data sets available for others..